Reg No.:_____ Name:_____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S8 (S) Examination September 2025 (2019 Scheme)

Course Code: ECT468

Course Name: RENEWABLE ENERGY SYSTEMS

Max. Marks: 100 **Duration: 3 Hours**

PART A

Answer all questions, each carries 3 marks. Marks

1		List any three merits and demerits of wind energy.	(3)
2		Give the importance of renewable energy in the current scenario.	(3)
3		Explain the term maximum power tracking (MPPT) in solar cell.	(3)
4		Explain the operating principle of tandem solar cell.	(3)
5		Give the different classifications of wind turbines.	(3)
6		How wind intensity is being evaluated?	(3)
7		Explain the need of islanding in power systems.	(3)
8		Explain the power quality management in power systems.	(3)
9		Discuss the main challenges of smart grid.	(3)
10		List out the different applications of SCADA systems.	(3)

PART B

Answer any one full question from each module, each carries 14 marks.

Module I

11	a)	Write short notes on a) Geothermal energy b) Biomass Energy	(8)
	b)	Explain any two renewable sources in India.	(6)

OR

12	a)	Explain the operation of hydropower systems with suitable diagrams.	(8)
	b)	Give the advantages and disadvantages of nuclear energy.	(6)

Module II

13	a)	Draw and explain the VI and PI characteristics of a solar cell with major electrical parameters?	(8)
	b)	Explain the working of a commercial solar cell.	(6)

OR

14	a)	Explain the different methods used for thin film device fabrication.	(14)

Module III

15 a) Explain the operation of horizontal axis wind turbines with its advantages. (8)

b) Explain the working of lift and drag type turbines. (6)

OR

16 a) Explain briefly about System Toroidal Rotor Amplifier Platform (TARP) and Wind amplified rotor platform (WARP). (8)

b) Explain the working of a wind energy conversion system (WECS) with a block diagram. (6)

Module IV

17 a) How power control and management system for grid integration is being done. (8)

b) Explain the main issues in the integration of converter based sources. (6)

OR

18 a) Explain basic schemes of electronic conversion system applicable to the generation of renewable energy. (8)

b) Explain the need for power control and management system for grid integration. (6)

Module V

19 a) Explain the different protocols used for smart metering. (8)

b) Differentiate between smart grid and conventional grid. (6)

OR

20 a) With the help of block diagram explain the working of a smart meter. (8)

b) Explain the different layers of SCADA system architecture. (6)

****